

Bisoprolol to prevent adverse cardiac events (PACE) in COPD: a multicentre, double-blind, randomised, controlled, phase 3 trial



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Summary

Background Although cardiovascular disease is common in patients with chronic obstructive pulmonary disease (COPD), the efficacy and safety of β blockers in reducing cardiac events and mortality is uncertain. This study was designed to assess whether the cardioselective β blocker, bisoprolol, improves cardiorespiratory outcomes when added to usual COPD care.

Methods This double-blind, randomised, controlled, phase 3 trial was done across 22 hospital and research institute sites selected by research experience and resources to conduct the study in Australia, India, New Zealand, and Sri Lanka. Participants aged 40–85 years with COPD, post-bronchodilator FEV₁ 30–70% predicted, and at least one COPD exacerbation in the previous 2 years were randomly assigned to receive bisoprolol (1·25–5 mg) or matched placebo, orally, once daily, for 2 years, with both groups continuing to receive usual COPD care. Randomisation was with a concealed, computer-generated sequence, stratified by site, smoking status, and previous diagnosis of cardiovascular disease requiring treatment. Participants, site personnel, and investigators remained masked to study treatment throughout the trial. The primary outcome was a composite of cardiac and respiratory effects, starting with the most important outcome (death), then cardiac or respiratory hospital admissions, exacerbations, quality-of-life measures, and FEV₁. Analysis was by intention-to-treat (ITT) in all randomly assigned patients, using a country-stratified win ratio. Adverse events were analysed from the ITT population using all available data. Missing data were not used to determine wins, but regarded as a tie on that level of the hierarchy. The trial was prospectively registered on ClinicalTrials.gov (NCT03917914), Clinical Trial Registry – India (CTRI/2020/08/027322), and the Sri Lanka Clinical Trials Registry (SLCTR/2021/033). The study is complete.

Findings Of 360 participants screened for eligibility between June 30, 2020, and March 20, 2023, 280 were randomly assigned to bisoprolol (n=143) or placebo (n=137), with 249 completing 2 years of follow-up. 233 patients (83%) were male and 47 (17%) were female; mean age was 68 years (SD 8). Mean post-bronchodilator FEV₁ was 45% (SD 11) predicted at baseline. According to the hierarchy of outcomes, bisoprolol was associated with better cardiorespiratory health for 3041 (45%) of 6763 comparisons and placebo for 3240 (48%), with 482 (7%) showing no difference, giving a win ratio of 0·95 (95% CI 0·72 to 1·25, p=0·72) and a net benefit of –2% (95% CI –15 to 10) with bisoprolol. No significant differences were seen between bisoprolol and placebo in all-cause mortality, cardiorespiratory hospitalisations, major adverse cardiac events, or moderate or severe COPD exacerbations. Additionally, no significant differences were seen in FEV₁, COPD symptoms, or quality of life, or adverse events. The most common adverse events were COPD exacerbations, occurring in 83 (58%) participants in the bisoprolol group and 87 (64%) in the placebo group. 15 (10%) participants in the bisoprolol group and 11 (8%) in the placebo group died. None of these deaths were attributed to the treatment.

Interpretation In patients with moderately severe COPD, treatment with bisoprolol made no difference to overall cardiorespiratory health, all-cause mortality, or serious cardiorespiratory events.

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Introduction

Patients with chronic obstructive pulmonary disease (COPD) have a high risk of cardiovascular disease.^{1,2} In patients with moderate COPD, the number of deaths due to cardiac causes can equal or exceed those from

respiratory disease.^{3–6} Although the reasons for the high, often undiagnosed, rates of cardiovascular disease in patients with COPD are not fully understood, they are not totally attributable to shared risk factors such as smoking, air pollution, or advancing age.¹ Low lung function itself

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Research in context

Evidence before this study

Cardiovascular disease is a leading cause of mortality in patients with chronic obstructive pulmonary disease (COPD), particularly after exacerbations. Despite this, cardiovascular disease is under-recognised in patients with COPD, who have often been excluded from clinical trials of cardiovascular treatments. We searched Ovid MEDLINE, containing MEDLINE and PRE-MEDLINE databases, from Jan 1, 2000, to June 30, 2025, using the terms “COPD”, “betablockers”, “bisoprolol” and “treatment outcomes”, with no language, journal, or study type restrictions. We identified 62 papers that covered COPD and β blockers and 52 papers that addressed COPD and bisoprolol. Although database analyses show that patients with COPD who have been prescribed β blockers have lower mortality and fewer exacerbations, concerns about their safety persist. Randomised controlled trials assessing the benefit and safety of β blockers in COPD had not been undertaken when the PACE study commenced. Subsequently, two trials have been published showing that neither metoprolol nor bisoprolol reduce COPD acute exacerbation rates. One of these studies suggested that bisoprolol was safe, but the other raised concerns about a higher risk of more severe COPD exacerbations with metoprolol.

Added value of this study

The PACE study is the first study of β blockers in COPD to use a composite cardiorespiratory endpoint. We believe that it is

also the first respiratory study to use the novel statistical approach of a win ratio analysis. This approach used a hierarchy of outcomes according to their clinical importance, including all-cause mortality, cardiorespiratory hospitalisations, COPD exacerbations, major cardiac events, quality of life, and FEV₁, to provide a holistic assessment of benefit and risk. We found that bisoprolol was not different to placebo in resulting in no overall benefit, nor was there any difference in secondary outcomes. However, we also found that bisoprolol was not associated with an excess of adverse events compared with placebo.

Implications of all the available evidence

This 2-year, randomised controlled trial extends the findings of two previous studies of β blockers in COPD indicating that, in the absence of specific cardiac indications, β blockers are not beneficial in patients with COPD. However, the findings are reassuring that the highly selective β_1 blocker bisoprolol has a good safety profile in patients with COPD who have a clear indication for β blocker treatment. We included a globally diverse ethnic population, suggesting that this finding is relevant across different ethnic origins and settings. The available evidence indicates that the highly cardioselective β blockers are safe, but do not improve cardiac or respiratory health in patients with moderately severe COPD.

predicts both fatal and non-fatal cardiovascular events, even in mild COPD.⁷ Exacerbations of COPD, in particular, are associated with a high risk for all-cause and cardiovascular death.^{8–10} Evidence of myocardial ischaemia or dysfunction at the time of an exacerbation is associated with a very high risk of death.¹¹

Despite the established evidence for high cardiovascular risk among patients with COPD, there is uncertainty as to how to reduce it. Standard treatments for COPD do not prevent cardiovascular events.^{5,6} Standard management of cardiovascular risks is recommended,¹² but the benefits of this approach are not proven, because patients with COPD have often been excluded from preventive cardiology trials. In particular, most of the key randomised controlled trials of β blockers for heart failure and other cardiac diseases excluded patients with COPD because of concern about their potential to worsen airflow obstruction.^{13,14} Even though most observational studies and meta-analyses indicate that patients with COPD do not have worsening lung function or respiratory symptoms when treated with cardioselective β_1 blockers, many practitioners remain reluctant to prescribe them and they are underused.^{15–17}

In addition to potential cardiovascular benefits, observational studies suggest that β blockers might reduce the risk of exacerbations of COPD and lower mortality.^{18,19} However, reports in 2019 and 2024 from

two randomised controlled trials to assess whether cardioselective β blockers reduce exacerbation rates found no benefit.^{20,21} In one of these, metoprolol was associated with a non-statistically significant increase in severe exacerbations, although this result was not seen in the second study, which used the more selective β_1 blocker, bisoprolol. However, neither of these previous trials were designed to address the cardioprotective effects of β blockers in COPD or their effect on overall cardiorespiratory health.

The Preventing Adverse Cardiac Events (PACE) in COPD trial aimed to assess the efficacy of bisoprolol for the prevention of death and adverse cardiorespiratory events in patients with moderately severe COPD.

Methods

Study design

Detailed methods have been described previously.²² In brief, we conducted a 2-year, multicentre, double-blind, randomised, placebo-controlled, phase 3 trial (PACE) at 22 hospital and research institute sites, which were selected by research experience and resources, in Australia, India, New Zealand, and Sri Lanka. The protocol (appendix pp 63–125) and the statistical analysis plan (appendix pp 27–62) are available in the appendix. The study was approved by the relevant ethics committees in each country (Health and Disability

Ethics Committee, New Zealand [2022AM9124]; Sydney Local Health District Human Research Ethics Committee, Australia [2019/PID09780]; Indian Council of Medical Research [2020–10090/F1]; SIDCER Ethics Review Committee, Sri Lanka [P/127/09/2021]). The study was monitored for safety and efficacy by an independent data and safety monitoring board, composed of senior clinicians and academics. The board reviewed masked allocated data for adverse events at regular intervals with 6-monthly meetings and received quarterly reports of serious adverse events. The trial was prospectively registered on ClinicalTrials.gov (NCT03917914), Clinical Trial Registry – India (CTRI/2020/08/027322), and the Sri Lanka Clinical Trials Registry (SLCTR/2021/033). The study is complete. The study steering committee included a consumer representative.

Participants

Eligible participants were aged 40–85 years, had been diagnosed with COPD according to the criteria of the Global Initiative for Chronic Obstructive Lung Disease (2019), and had a COPD exacerbation treated with oral corticosteroids or antibiotics, or both, within the previous 24 months (due to a lower incidence of COPD exacerbations during the COVID-19 pandemic, the timeframe for a COPD exacerbation was extended from 12 to 24 months in February, 2021). FEV₁ was required to be between 30% and 70% of predicted, with an FEV₁/FVC ratio less than 0·7 post-bronchodilator. Maintenance oral corticosteroids were permitted if at a stable dose of no greater than 10 mg prednisone per day for 4 weeks. In New Zealand, an additional ethics committee requirement was for patients to have a history of cardiovascular disease (heart failure, ischaemic heart disease, arrhythmia, or hypertension).

Exclusion criteria were: concurrent β blocker treatment, resting heart rate less than 60 beats per minute (bpm), systolic blood pressure less than 100 mm Hg, unstable left heart failure in the previous 12 weeks, clinically significant pulmonary hypertension considered to be a contraindication to β blockers, severe peripheral vascular disease, second-degree or third-degree heart block, long-term oxygen therapy, resting oxygen saturation less than 90%, lower respiratory tract infection or exacerbation of COPD within the previous 4 weeks, coexisting asthma, bronchodilator response greater than 400 mL on two occasions, clinical instability following a major adverse cardiovascular event in the previous 12 weeks, expected survival less than 12 months or coexisting illness precluding participation as judged by the investigator, end-stage liver disease, pregnancy or not using adequate contraception in women of childbearing age, and a high chance of not following the study protocol in the opinion of the investigators.

Participants who provided written informed consent before enrolment in the study and met eligibility

criteria were given a single 1·25 mg test dose of bisoprolol. Blood pressure was measured at 30 min, 1 h, and 2 h, and spirometry was repeated at 1 h and 2 h after the test dose. Participants were excluded from further participation if their systolic blood pressure fell below 100 mm Hg at 1 h or 2 h or FEV₁ fell by at least 200 mL and at least 12% from the pre-dose value at 2 h. Sex was self-reported as male, female, or other. Ethnicity was also self-reported.

Randomisation and masking

Eligible participants with COPD were randomly assigned 1:1 to receive bisoprolol plus usual care (intervention) or matching placebo plus usual care (control). The randomisation list was concealed and generated centrally by a computer algorithm using a random combination of four by six permuted blocks, stratified by site, smoking status, and previous diagnosis of cardiovascular disease requiring treatment. The randomisation sequence was generated by an unmasked statistician. The study site investigators recruited and enrolled the participants. Allocation to treatment groups was done via a web-based system with the randomisation lists loaded and masked to all users. The unmasked statistician had no involvement in recruiting or randomly assigning participants. Bisoprolol and matched placebo were labelled using concealed kit numbers. Participants, site personnel, and investigators remained masked to the study treatment throughout the trial.

Procedures

The initial dose of study medication was 1·25 mg bisoprolol (half of a 2·5 mg tablet) orally, once daily, or matching placebo, and was titrated according to a standard protocol: at the first follow-up visit after 1 month (visit 2), the dose was increased to 2·5 mg (one tablet) daily, and was further increased to 5 mg (two tablets) daily at the next follow-up (visit 3), 3 months after baseline. The investigator could use their discretion to not increase, or to reduce the dose, if the participant, at either visit, had a fall in FEV₁ from baseline ($\geq 12\%$ and ≥ 200 mL), systolic blood pressure less than 100 mm Hg, heart rate less than 60 bpm, symptomatic postural hypotension, increased wheeze or chest tightness, marked exercise-induced fatigue, or symptoms of peripheral arterial insufficiency.

Participants visited the study centres at 0 (baseline and randomisation), 1, 3, 6, 12, 18, and 24 months. Pre-bronchodilator spirometry was done at baseline. Post-bronchodilator spirometry, blood pressure, heart rate, participant-reported adverse events, and changes in concomitant medications were recorded at all visits. Electrocardiograms were done at baseline, and at 1, 3, 12, and 24 months. 6-min walk tests were done at baseline and at 12 months at self-selected sites with the capacity and resources to do them. The St George Respiratory Questionnaire (SGRQ),²³ EuroQoL Group 5-Dimension

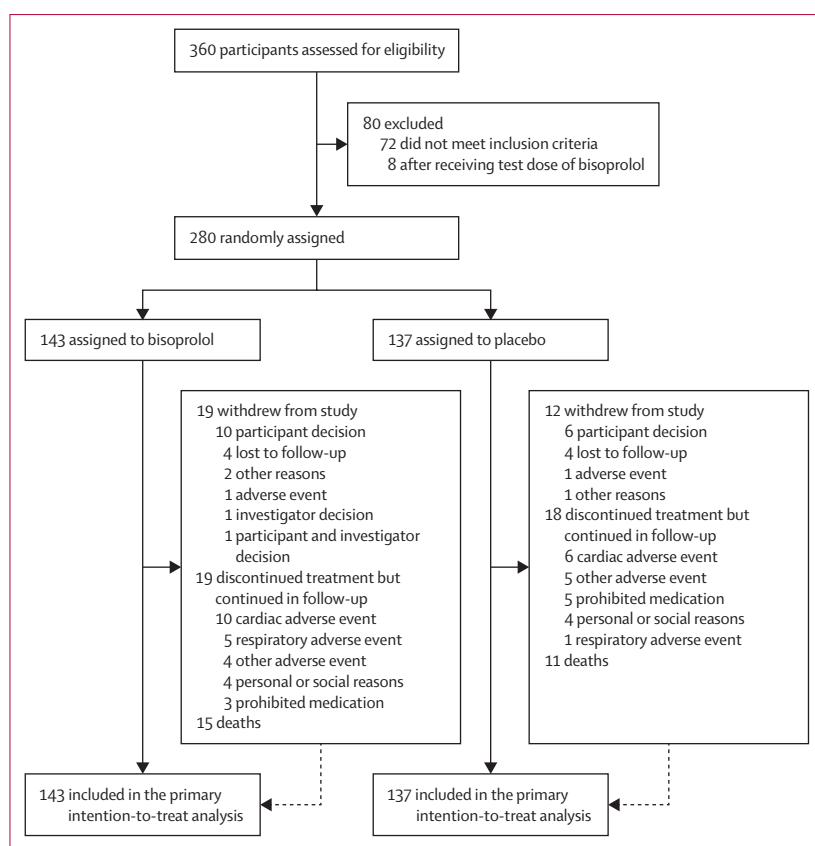


Figure 1: Trial profile

self-report questionnaire (EQ-5D-5L),²⁴ and COPD Assessment Test (CAT)²⁵ were completed at baseline, and at 6, 12, 18, and 24 months. Telephone calls to participants to record adverse events and adherence were made by site coordinators (typically study nurses) at 2 weeks, 8 weeks, and approximately every 8 weeks between clinic visits. Adverse events were defined by the protocol and were self-reported at the clinic visits and telephone calls to the participants. Site investigators were responsible for determining causality of adverse events according to the criteria of the International Council for Harmonisation Good Clinical Practice guideline, specified in the manual of procedures. Adherence to treatment was assessed by tablet counts at clinic visits.

All participants continued to receive usual best-practice COPD care according to their treating physician, and local and national guidelines. They also received usual care for other medical conditions, but were asked to avoid other β blockers, non-dihydropyridine calcium channel blockers (with the exception of diltiazem at the investigator's discretion), and class 1a or 1c anti-arrhythmic agents. If it was deemed clinically necessary to use one of these medications, participants were instructed to cease the randomly assigned intervention before starting the medication, but were requested to continue with follow-up visits.

Outcomes

Our initial study protocol had a composite cardiorespiratory primary outcome of time to the first event including any of: all-cause mortality or hospitalisation for a COPD exacerbation, non-fatal myocardial infarction, non-fatal stroke, cardiac ischaemia, arrhythmia, or heart failure. Secondary outcomes were annualised COPD exacerbation rates, time to first moderate or severe COPD exacerbation, annualised hospitalisation rates for respiratory and cardiac causes, total number of cardiac events, quality of life (SGRQ, EQ-5D-5L, and CAT symptom scores), spirometry, and 6-min walk distance. Due to disruptions to recruitment and lower exacerbation rates during the COVID-19 pandemic, it became clear that we would not meet the required sample size for the primary outcome and we decided to use a win ratio approach, with a composite of the primary and secondary outcomes, which requires a smaller sample size. This change was lodged with the relevant trial registration bodies and published before the randomisation code was broken and before any outcome analyses were undertaken.²²

The win ratio compares the outcomes for every individual in the active (bisoprolol) treatment group with the outcomes of every individual in the control (placebo) group. Outcomes are ordered in a hierarchy, starting with the most clinically important. At each step of the hierarchy, the comparison evaluates which treatment had the better outcome (the winner).²⁶ When outcomes do not differ between the treatments (a tie), the comparison then uses the next outcome. Once a winner has been established, the remainder of the hierarchy is not considered. Because of the large number of comparisons and the likelihood that a win can be decided for most comparisons, this approach has the potential to improve the power of the study as well as incorporating multiple outcomes, providing a comprehensive and holistic assessment of treatment effect.

The hierarchy of outcomes is shown in the appendix (p 2). The highest level of the hierarchy was time to death from any cause (ie, if one or both participants in a comparison died during the study, the treatment taken by the one who survived the longest was the winner). For comparisons in which both participants survived (ties), the next level of the hierarchy (number of cardiorespiratory hospitalisations) was used, and so on.

The cause of all cardiac and respiratory hospitalisations and all deaths was initially attributed by site investigators and then independently adjudicated by a PACE investigator from another site for respiratory events or by advanced cardiology trainees for cardiac events. If this opinion differed from the site investigator, the final adjudication was made by an independent adjudication committee provided with all relevant documentation. Exacerbations of COPD were graded as moderate if they required treatment with oral corticosteroids, antibiotics, or both, and severe if they also required hospital

See Online for appendix

admission. Mild exacerbations were also recorded as symptomatic events that were treated only with increased inhaled or non-steroid oral medications.

Statistical analysis

The trial was initially powered to detect a hazard ratio of 0.75 in time to first composite event, using survival analysis, and required a sample size of 1164 with 514 participants having at least one event. Due to recruitment challenges, only 280 patients were randomly assigned to treatment and the primary outcome was changed to a win ratio analysis. Assuming a win ratio of 1.5 (54% of wins with bisoprolol) and 10% of ties, this approach had 76% power. Although the proportion of wins was difficult to anticipate, partly due to the absence of previous win ratio analyses in this field, we felt that 54% of wins in the bisoprolol group was plausible and would indicate a worthwhile effect. Given the number of steps proposed for the win ratio analysis, we believed that 10% of ties was a conservative estimate.

The statistical analysis plan was developed before any unmasking (appendix pp 27–62). The primary analysis used the win ratio hierarchy in the appendix (p 2). Each participant in the bisoprolol group was paired with each participant in the placebo group. The number of wins for each group was used to calculate the win ratio. We calculated CIs and p values using the approach described by Dong and colleagues.²⁷ Because of expected differences in population characteristics and health-care delivery between the four countries involved (eg, the threshold at which a patient is admitted to hospital for a COPD exacerbation is likely to differ between countries), the win ratio analysis was stratified by country so that participants were compared only within countries and not with participants in other countries.

The primary analysis was by intention-to-treat, including all participants who were randomly assigned to treatment. A supplemental per-protocol analysis excluded those who stopped study drug or withdrew, those who did not follow the dose titration schedule, and those who did not adhere to the study medication or visit schedule. All randomly assigned participants were included in the safety analysis. Participants were encouraged to remain in the study even if they had stopped taking the study treatment. Participants who withdrew or were lost to follow-up were compared with other participants only for the time for which we had data. For example, if a participant withdrew after 1 year, only the data for the first year of follow-up were used for each comparison.

Additional prespecified analyses included subgroup analyses of the win ratio by sex, age group, continent, and pre-existing cardiovascular disease. Kaplan–Meier survival analyses of time to death, hospitalisation for respiratory and cardiac causes, moderate or severe COPD exacerbations, major adverse cardiac events, and all serious and non-serious adverse events were

	Bisoprolol (n=143)	Placebo (n=137)	Overall (N=280)
Age, years	68.3 (8.0)	66.9 (7.3)	67.6 (7.7)
Sex			
Male	119 (83%)	114 (83%)	233 (83%)
Female	24 (17%)	23 (17%)	47 (17%)
Post-bronchodilator FEV ₁ , L	1.072 (0.314)	1.135 (0.339)	1.103 (0.328)
Post-bronchodilator FEV ₁ , % predicted	44 (11)	46 (12)	45 (11)
Systolic blood pressure, mm Hg	131.6 (15.3)	131.0 (14.7)	131.3 (15.0)
Diastolic blood pressure, mm Hg	80.9 (9.9)	80.7 (9.2)	80.8 (9.6)
Heart rate, beats per minute	82.5 (11.0)	82.2 (9.7)	82.3 (10.4)
BMI, kg/m ²	22.6 (6.2)	22.9 (7.4)	22.7 (6.8)
Ethnic group			
European ancestry	34 (24%)	24 (18%)	58 (21%)
South Asian	103 (72%)	104 (76%)	207 (74%)
Other Asian	3 (2%)	2 (1%)	5 (2%)
Australian Aboriginal	2 (1%)	1 (1%)	3 (1%)
Māori	1 (1%)	4 (3%)	5 (2%)
Other	0	2 (1%)	2 (1%)
Smoking history			
Never smoked	12 (8%)	10 (7%)	22 (8%)
Current smoker	17 (12%)	18 (13%)	35 (13%)
Past smoker	114 (80%)	109 (80%)	223 (80%)
Medical history			
Surgery on lungs	7 (5%)	5 (4%)	12 (4%)
Sleep apnoea	7 (5%)	11 (8%)	18 (6%)
Diabetes	17 (12%)	21 (15%)	38 (14%)
Coronary heart disease	17 (12%)	15 (11%)	32 (11%)
Heart failure	9 (6%)	8 (6%)	17 (6%)
Hypertension	53 (37%)	51 (37%)	104 (37%)
Other heart conditions	10 (7%)	6 (4%)	16 (6%)
Stroke	3 (2%)	2 (1%)	5 (2%)
Osteoporosis	13 (9%)	8 (6%)	21 (8%)
Other significant medical conditions	67 (47%)	65 (47%)	132 (47%)
COPD medication			
Corticosteroids, oral	0	0	0
Corticosteroids, inhaled	95 (66%)	104 (76%)	199 (71%)
Long-acting β_2 agonists	103 (72%)	108 (79%)	211 (75%)
Long-acting muscarinic antagonists	65 (45%)	75 (55%)	140 (50%)
Short-acting β_2 agonists	91 (64%)	87 (64%)	178 (64%)
Short-acting muscarinic agonists	22 (15%)	20 (15%)	42 (15%)
Theophylline	20 (14%)	15 (11%)	35 (13%)
Mucolytics	1 (1%)	4 (3%)	5 (2%)
Natural therapies and vitamins	8 (6%)	7 (5%)	15 (5%)

Data are n (%) or mean (SD). COPD=chronic obstructive pulmonary disease.

Table 1: Baseline characteristics

undertaken. Cox models with adjustments for cardiovascular history, and sites as random effects, were used to test differences in survival, and estimate hazard ratios and 95% CIs. A sensitivity analysis for the composite cardiopulmonary outcome adjusted for age, sex, and smoking status. Annual rates of COPD exacerbations were analysed by Poisson regression with an over-dispersion parameter, adjusted for cardiovascular

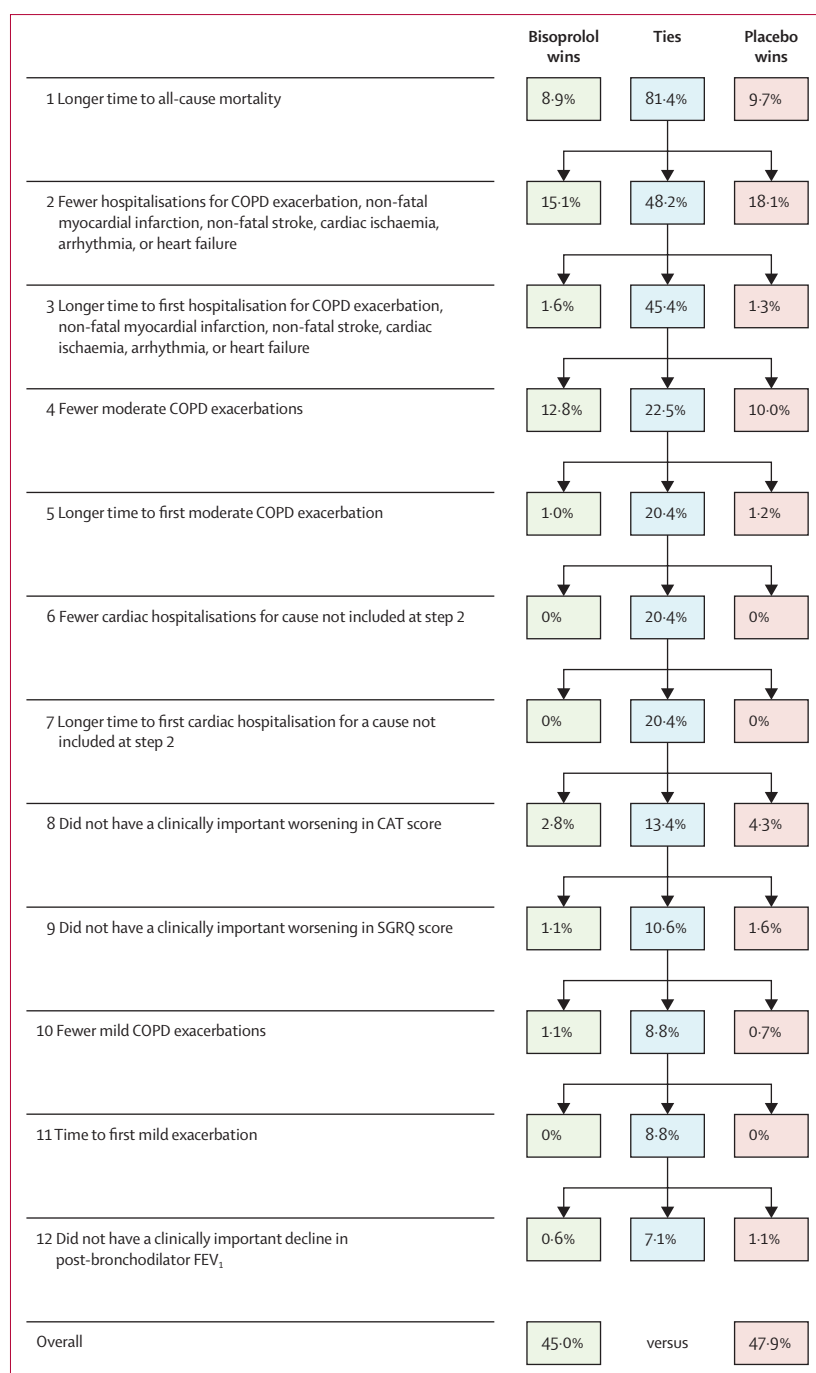


Figure 2: Flow diagram of win ratio decisions

Overall stratified win ratio of 0.95 (95% CI 0.72–1.25; $p=0.72$). CAT=COPD Assessment Test. COPD=chronic obstructive pulmonary disease. SGRQ=St George Respiratory Questionnaire. For the number of pair comparisons involved in each decision, see appendix p 21.

history and smoking status and reported as risk ratios and 95% CI. Continuous variables (spirometry measurements, vital signs, CAT scores, SGRQ, and EQ-5D-5L visual analogue score) were analysed using linear mixed models including the randomised arm, visit as a categorical effect, and their interaction.

Correlations between individual measurements were modelled through random effects. The effect of the intervention was estimated as the overall mean difference and 95% CI, weighted to reflect the uneven spacing of visits. 6-min walk tests were analysed using linear mixed models with adjustments for cardiovascular history, smoking status, and baseline measurement, with sites as random effects. For continuous outcomes, normality assumptions were assessed through quantile–quantile plots and histograms of residuals. For all previously described models, the p values were calculated using Wald tests. The proportions of patients who had at least one adverse event were compared using Fisher's exact tests.

Analyses were conducted in R (version 4.5.0) using the WINS package for the win ratio. Two-sided p values <0.05 were regarded as statistically significant.

Role of the funding source

The study funders had no role in the study design, data collection, data analysis, data interpretation, or writing of the report.

Results

Of 360 participants screened between June 30, 2020, and March 20, 2023, 280 participants from 22 study sites were randomly assigned to treatment. 143 patients were given bisoprolol and 137 placebo, with 249 (89%) completing the 2-year follow-up (figure 1). There were 45 participants recruited in Australia, 70 in India, 25 in New Zealand, and 140 in Sri Lanka (appendix p 5). Baseline characteristics are shown in table 1. Most participants were male (233 of 280 [83%]) and the majority (207 [74%]) were of South Asian ethnicity, with a further 58 (21%) reporting European ethnicity. The doses of bisoprolol and placebo achieved are shown in the appendix (p 7): by 6 months, 58 (50%) of 117 participants continuing in the study were taking the full 5 mg dose of bisoprolol, 40 (34%) were taking 2.5 mg daily, and 19 (16%) remained on the lowest 1.25 mg dose. For placebo, the equivalent participant numbers were 69 (61%) of 113 participants, 35 (31%), and nine (8%), respectively. Tablet counts indicated a median adherence of 94% (IQR 0.81–1.00) for bisoprolol and 96% (0.85–1.00) for placebo, although 11 (8.5%) of 130 participants had less than 50% adherence in the bisoprolol group and 4 (3.2%) of 125 had less than 50% adherence in the placebo group. Adherence could not be estimated for 13 of 143 participants in the bisoprolol group and 12 of 137 in the placebo group due to missing data (appendix p 12). 37 participants withdrew from treatment during the study: 19 from the bisoprolol group, and 18 from the placebo group. Overall, nine (3%) participants were started on an open-label β blocker for clinical reasons by the end of the study: three (2%) participants in the bisoprolol group and six (4%) in the placebo group.

	Bisoprolol wins	Placebo wins	Ties	Estimate (95% CI)	p value
Primary composite*	3041 (45.0%)	3240 (47.9%)	482 (7.1%)	0.95 (0.72–1.25)	0.72
Individual components†					
1 Longer time to all-cause mortality	601 (8.9%)	658 (9.7%)	5504 (81.4%)	0.86 (0.39–1.94)	0.73
2 Fewer hospitalisations for COPD exacerbation, non-fatal myocardial infarction, non-fatal stroke, cardiac ischaemia, arrhythmia, or heart failure	1225 (18.1%)	1494 (22.1%)	4044 (59.8%)	0.89 (0.54–1.47)	0.66
3 Longer time to first hospitalisation for COPD exacerbation, non-fatal myocardial infarction, non-fatal stroke, cardiac ischaemia, arrhythmia, or heart failure	1337 (19.8%)	1603 (23.7%)	3823 (56.5%)	0.92 (0.57–1.47)	0.72
4 Fewer moderate COPD exacerbations	1763 (26.1%)	1728 (25.6%)	3272 (48.4%)	0.97 (0.66–1.43)	0.87
5 Longer time to first moderate COPD exacerbation	1855 (27.4%)	1994 (29.5%)	2914 (43.1%)	0.94 (0.65–1.34)	0.72
6 Fewer cardiac hospitalisations for cause not included at step 2	0	17 (0.3%)	6746 (99.7%)	NA‡	NA‡
7 Longer time to first cardiac hospitalisation for a cause not included at step 2	0	17 (0.3%)	6746 (99.7%)	NA‡	NA‡
8 Did not have a clinically important worsening in CAT score	994 (14.7%)	1122 (16.6%)	4647 (68.7%)	0.95 (0.54–1.64)	0.84
9 Did not have a clinically important worsening in SGRQ score	684 (10.1%)	1028 (15.2%)	5051 (74.7%)	0.87 (0.48–1.58)	0.65
10 Fewer mild COPD exacerbations	1164 (17.2%)	834 (12.3%)	4765 (70.5%)	1.45 (0.76–2.76)	0.26
11 Time to first mild exacerbation	1164 (17.2%)	915 (13.5%)	4684 (69.3%)	1.33 (0.71–2.47)	0.37
12 Did not have a clinically important decline in post-bronchodilator FEV ₁	971 (14.4%)	1134 (16.8%)	4658 (68.9%)	0.88 (0.51–1.52)	0.65

Data are n (%) of 6763 (the total number of pair comparisons after stratification by country) unless indicated otherwise. CAT=COPD Assessment Test. COPD=chronic obstructive pulmonary disease. SGRQ=St George Respiratory Questionnaire. *Every participant randomised to the bisoprolol group has been paired with every participant randomised to the placebo group within the same country. Within each pair, we compared the outcomes in a hierarchical fashion until a winner was determined. If no winner could be declared after comparing all outcomes, the pair was considered a tie. The analysis has been stratified by country using the Mantel-Haenszel weighting method. †The win-ratio has been performed independently for each component and stratified by country, including all patients with non-missing outcomes. ‡NA=not applicable: because of zero events in one group, the ratio and p value are not calculated.

Table 2: Win ratio analysis of the primary outcome and its components

Heart rate fell from a mean of 82.5 bpm (SD 11) at baseline to 76.5 bpm (10.9) at month 24 in the bisoprolol group, and remained constant in the placebo group (82.2 bpm [9.7] at baseline and 81.6 bpm [10.4] at month 24). No difference was observed between groups in systolic blood pressure, but diastolic blood pressure was numerically lower in the bisoprolol group between week 4 and month 12 (appendix p 9 and pp 16–17).

Win ratio comparisons favoured placebo for 3240 (48%) comparisons and bisoprolol for 3041 (45%) with 482 (7%) of comparisons being ties, giving a win ratio of 0.95 (95% CI 0.72 to 1.25; $p=0.72$). The net benefit of bisoprolol was –2% (95% CI –15 to 10). The breakdown of where the wins were decided is shown in figure 2 and in the appendix (p 21). The comparisons of each component of the win ratio (without using the hierarchy) are shown in table 2. Numerically, the percentage wins favoured placebo across most outcomes except for the number of moderate COPD exacerbations, fewer mild COPD exacerbations, and longer time to first mild exacerbation, but none of these comparisons showed a statistically significant difference favouring either treatment group. Findings from the per-protocol analysis were similar, with a win ratio of 0.96 (0.62–1.48; $p=0.86$).

Survival analyses of time to death, hospitalisation for cardiac or respiratory causes, or moderate or severe exacerbations showed no clinically or statistically significant differences between treatment groups for the

primary composite measure or for any of the components of this measure (figure 3). No differences were observed in quality of life or symptom scores (SGRQ, EQ-5D-5L, CAT, appendix pp 10–11). At 24 months, the mean total SGRQ scores were: bisoprolol 38.7 (SD 19.3), placebo 37.1 (18.0); mean EQ-5D-5L utility scores were bisoprolol 0.85 (SD 0.17), placebo 0.87 (0.14), mean CAT scores were bisoprolol 17.6 (SD 6.5), placebo 17.0 (6.8). Mean percent predicted FEV₁ values at 24 months were 45.1 (SD 14.9) for bisoprolol and 46.5 (15.5) for placebo (appendix p 8). No differences were observed in the number of serious or total adverse events (table 3). Most (90%) of participants in both bisoprolol and placebo groups had at least one adverse event, most of which were judged to be mild. The most common adverse events were COPD exacerbations requiring antibiotic or corticosteroid treatment, or both. These occurred in 83 (58%) of 143 participants in the bisoprolol group and 87 (64%) of 137 in the placebo group. There were no treatment-related deaths. Annualised COPD exacerbation rates during the study were 0.95 (95% CI 0.73–1.23) for placebo and 0.89 (0.69–1.16) for bisoprolol. Subgroup analyses by age group, sex, continent, and cardiovascular history did not support effect modification by any of these characteristics (appendix p 19).

The mean 6-min walk distances after 12 months were 363 m (SD 97) in the bisoprolol group and 352 m (88) in the placebo group: a mean difference of 12 m (95% CI

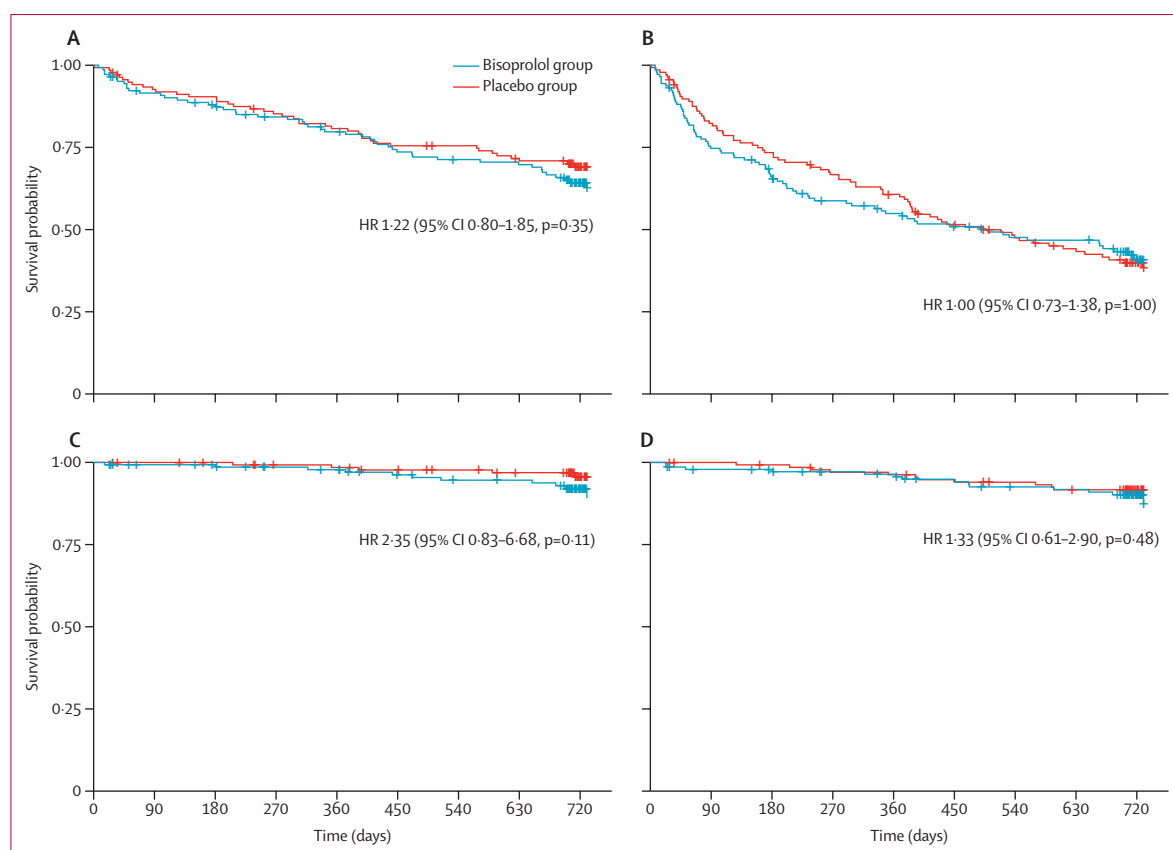


Figure 3: Kaplan-Meier survival plots for time to major study outcomes

Composite primary outcome of all-cause mortality or hospitalisation for COPD exacerbation, non-fatal myocardial infarction, non-fatal stroke, cardiac ischaemia, arrhythmia, or heart failure (A); moderately severe COPD exacerbations (B); major adverse cardiovascular events (C); and all-cause mortality (D). Hazard ratios and 95% CIs are estimated using a Cox model adjusted on cardiovascular disease history. The sites have been included as random effects. HR=hazard ratio.

–10 to 34; $p=0.29$) adjusted for baseline walk distance (appendix pp 10, 25).

Discussion

This randomised controlled trial found no evidence that bisoprolol improves cardiovascular or respiratory outcomes in patients with moderately severe COPD. Importantly, we also found no evidence that bisoprolol was harmful. There was little evidence that bisoprolol and placebo differed on any of the outcomes, including hospitalisations, exacerbations, quality of life, lung function, exercise tolerance, and adverse effects.

The findings add to those from the Bisoprolol In COPD Study (BICS),²¹ which also found no evidence that bisoprolol prevented COPD exacerbations or any evidence of a greater number of adverse effects compared with placebo. Our findings are reassuring in light of the concerns raised by the BLOCK-COPD study, which was stopped early due to futility, with no significant between-group difference in time to first COPD exacerbation, but raised safety concerns with higher numbers of severe exacerbations in the metoprolol arm compared with placebo.²⁰ A potential reason for this difference between

BLOCK-COPD and the findings of BICS and the present study is the use of metoprolol in the BLOCK-COPD study, which is less cardioselective than bisoprolol.²⁸ It is also possible that differences between the study populations contributed to the different findings: participants in BLOCK-COPD had slightly worse lung function and CAT scores, and high rates of domiciliary oxygen use (an exclusion criterion in our study). We also recruited participants with a lower pre-trial exacerbation rate than the BICS or BLOCK-COPD studies and the annualised exacerbation rates during the study were lower than those observed in BLOCK-COPD and BICS.²⁹ Despite this, 170 (61%) participants had COPD exacerbations requiring treatment with antibiotics or corticosteroids, or both, during the PACE study.

Importantly, this is the first randomised controlled trial of a β blocker in COPD to use a composite of cardiac and respiratory effects as the primary outcome. Both BLOCK-COPD and BICS were designed to assess whether β blockers reduced the risk of COPD exacerbations, which is a less biologically plausible effect compared with improved cardiac outcomes. Although they also reported no reduction in cardiovascular outcomes with

	Bisoprolol (n=143)		Placebo (n=137)		Overall (n=280)		p value*
	Events, n	Participants, n (%)	Events, n	Participants, n (%)	Events, n	Participants, n (%)	
All adverse events	815	128 (90%)	842	123 (90%)	1657	251 (90%)	0·94
Mild	425	107 (75%)	446	109 (80%)	871	216 (77%)	..
Moderate	269	78 (55%)	295	77 (56%)	564	155 (55%)	..
Severe	121	62 (43%)	101	55 (40%)	222	117 (42%)	..
Serious adverse events	106	62 (43%)	93	55 (40%)	199	117 (42%)	0·59
Resulted in death	15	15 (10%)	11	11 (8%)	26	26 (9%)	..
Life threatening	1	1 (1%)	0	0	1	1 (<1%)	..
Required prolonged hospitalisation	89	50 (35%)	79	46 (34%)	168	96 (34%)	..
Medically significant	4	4 (3%)	4	3 (2%)	8	7 (3%)	..
Cardiovascular events	13	12 (8%)	5	5 (4%)	18	17 (6%)	0·10
Cardiovascular death	10	9 (6%)	4	4 (3%)	14	13 (5%)	..
Non-fatal myocardial infarction	2	2 (1%)	0	0	2	2 (1%)	..
Non-fatal stroke	1	1 (1%)	1	1 (1%)	2	2 (1%)	..
COPD exacerbations	232	83 (58%)	246	87 (64%)	478	170 (61%)	0·35
Took antibiotics only	56	39 (27%)	48	34 (25%)	104	73 (26%)	..
Took corticosteroids only	36	28 (20%)	44	27 (20%)	80	55 (20%)	..
Took both antibiotics and corticosteroids	109	51 (36%)	108	50 (36%)	217	101 (36%)	..

The number of events is the total number of adverse events recorded. The number of participants is the number (%) of participants having at least one event. *p values have been calculated using Fisher's exact test to compare the overall proportion of patients having events between the two randomised groups.

Table 3: Summary of adverse events

β blockers, both studies excluded participants with established cardiac indications for these drugs, reducing the likelihood of observing an effect. We did not exclude participants with indications for β blockers (providing that they were not already taking them), and many participants had potential indications, with 104 (37%) participants having hypertension, 32 (11%) having coronary disease, and 17 (6%) having had heart failure. When the subgroups with and without a history of cardiovascular disease or hypertension were analysed separately, we did not observe an improvement in cardiorespiratory outcomes in either group (appendix p 19), although these analyses had limited power.

We believe that this is one of the first randomised, double-blind, placebo-controlled respiratory trials to use the win ratio. Although win ratio analyses have been used in cardiology studies for some time,²⁶ this approach is new in respiratory medicine. Our decision to use the win ratio was initially a pragmatic response to the impossibility of recruiting our target sample in the setting of the COVID-19 pandemic. The win ratio provides a more powerful analysis than the traditional approach of comparing a primary outcome and the required sample size is much smaller.²⁶ Additionally, the win ratio also provided a more holistic assessment of cardiorespiratory health, including both the number and timing of the primary and secondary outcome events according to their hierarchy of clinical importance, including quality-of-life scores and lung function. Arguably, this approach gets closer to providing clinicians and patients with what they need to know to decide whether a treatment is likely

to be of overall benefit. The investigators decided on the hierarchy before the analysis started, but differences in opinion on the order of clinical importance could potentially influence the overall finding. However, the fact that none of the steps in the hierarchy showed meaningful differences between placebo or bisoprolol (table 2) suggests that our null finding is robust to the ordering of the hierarchy.

Placebo was the winning treatment in 48% of comparisons, compared with 45% for bisoprolol, giving an overall win ratio of 0·95. The 95% CI for the ratio ranged from 0·72 to 1·25, indicating that the true win ratio is likely to be somewhere in this range. Our original study and analysis plan was designed to detect a 25% reduction in the composite primary outcome of deaths, cardiorespiratory admissions, and major adverse cardiac events with bisoprolol.²² Although the outcome measure for the win ratio is somewhat different, a 25% improvement in outcomes is at the limit of the plausible range suggested by the 95% CI. The first three levels of the win ratio hierarchy encompass our original primary outcome. Using just these three levels, 25% of comparisons were wins for bisoprolol and 29% were wins for placebo, giving a win ratio 0·91 (95% CI 0·60–1·39). Together with the negative findings of the BLOCK-COPD and BICS studies, these findings indicate that cardioselective β blockers do not improve outcomes in unselected patients with moderately severe COPD. The placebo group numerically won more comparisons than bisoprolol and it is plausible that the true win ratio is at the lower end of the 95% CI and that we were underpowered to detect harm. However, the

observed difference was small and neither this study nor BICS observed a significant excess of adverse effects with bisoprolol over placebo. It is also important to note that we found no evidence that bisoprolol treatment impaired exercise tolerance: 6-min walk distances were slightly, but not significantly, greater with bisoprolol (appendix pp 10, 25).

The main limitation of this study was a smaller sample size than originally planned, as a consequence of the COVID-19 pandemic's effect on study recruitment and the funding timeframe, which prevented further extension of recruitment. This reduced the power of the study to assess the effects of bisoprolol on the original combined primary outcome of mortality and major adverse cardiorespiratory events using a conventional statistical approach, although use of the win ratio mitigated this. Even with the win ratio approach, the study had slightly less than the conventional 80% power. Although only half of participants reached the target dose of 5 mg per day for bisoprolol and 14 (13%) of 111 participants in the bisoprolol group who completed the study stopped taking bisoprolol during the 2-year trial, the number of participants achieving the 5 mg per day bisoprolol dose is higher than in the BICS study, with fewer stopping the medication altogether. Further, the reduction in heart rate was similar to the BLOCK-COPD study of metoprolol, which indicates that most participants had adequate β blockade.^{20,21} It is notable that, even among the placebo group, only 69 (61%) participants were titrated to the full dose of two tablets a day. This finding likely reflects the morbidity of the population and the reluctance of the treating physicians to prescribe high doses of β blockers in these patients. Overall adherence was good, with a median value of 95%, but a small number (15 [6%]) of participants had less than 50% adherence and adherence could not be estimated for 25 (9%) patients.

Similar to both the BLOCK-COPD and BICS studies, the number of cardiovascular events during the study was smaller than anticipated, with only 17 (6%) participants having a major adverse cardiac event. This result is lower than reported by earlier randomised controlled trials of other treatments in COPD and was far lower than the number of participants who had moderate or severe COPD exacerbations (170 [61%]). A pilot study found that it was not feasible to recruit patients and start β blocker treatment at the time of an acute exacerbation, when the risk of cardiovascular events is highest.³⁰ Hence we recruited stable patients with a history of exacerbations because this predicts the risk for future exacerbations, and we followed up participants for 2 years, with rigorous processes to increase the chance of recording cardiorespiratory outcomes.¹²

We decided to use the most selective β blocker clinically available, and the finding that bisoprolol did not increase adverse COPD events should not be generalised to less

selective β blockers. We also gave a test dose of 1.25 mg bisoprolol to check safety before randomisation. Administration of a test dose is unlikely to be done in clinical practice, but we found that 95% of participants tolerated the test dose, with only nine (3%) excluded because of a fall in FEV₁ and six (2%) because of a fall in blood pressure.³¹

Most participants in the study were male. This likely reflects the higher prevalence of diagnosed COPD among men than women in India and Sri Lanka, but may limit the generalisability of the findings to women with COPD.

This study also has a number of strengths that complement earlier research. We recruited participants from multiple countries, including a high proportion from India and Sri Lanka, enhancing the generalisability of the findings to a broader ethnic mix and health-care systems. Participants with and without cardiovascular indications for β blockers were included, providing that β blockers had not already been started, further enhancing generalisability. The study was also conducted over 2 years, which is twice as long as the other randomised controlled trials on this topic, and provides data for the longer-term safety and efficacy of cardioselective β blockers in patients with COPD.

Our findings indicate that β blockers have no benefit for primary prevention of cardiac or respiratory events in patients with COPD without cardiovascular indications. However, the absence of an elevated risk of adverse events shows that that highly selective β_1 blockers are well tolerated in patients with COPD if there is a clear indication for a β blocker. Although the high risk of cardiovascular disease in patients with COPD is well established from observational studies, particularly in the setting of an acute exacerbation, the low rate of major cardiovascular events in this and previous randomised controlled trials of β blockers suggests that the absolute risk of these events should be re-evaluated. Future randomised controlled trials in respiratory medicine should also consider using the win ratio approach to analysis. This method is more efficient than traditional approaches and allows a more comprehensive assessment of health, including mixing subjective quality-of-life outcomes with objective event-based measures.

To conclude, the highly selective β blocker bisoprolol has a good safety profile but does not improve cardiac or respiratory health in patients with moderately severe COPD.

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Contributors

CRJ, CLC, RB, JPW, and RJH conceived the study. CRJ, AM, CLC, and RJH were primarily responsible for developing the study concept and design. CLC, JPW, VMM, CCD, IAY, CSF, BC, CDR, HAD, ANA, and RJH actively recruited participants at sites. All authors participated in finalisation of the protocol and ethics submissions, with GSH providing cardiac expertise. CPS was the consumer representative on the study steering group. CRJ, AM, CLC, and RJH were responsible for study oversight and conduct over 4–5 years. SG was primarily responsible for study monitoring and adjudication. LB drafted the initial statistical analysis plan, advised on the win ratio, and wrote and finalised the revised statistical analysis plan with input from CRJ, AM, CLC, and RJH, and final review by all authors. LB, AD, and CG verified the data and were responsible for the analysis; CRJ, AM, CLC, and RJH were primarily responsible for data interpretation. RJH wrote the first draft of the manuscript and CRJ, AM, CLC, and RJH revised and refined subsequent drafts. All authors were responsible for critical revision of the manuscript for important intellectual content, and contributed and approved the submitted manuscript. All authors had full access to all the data in the study and no author was precluded from accessing study data. The authors accept responsibility to submit for publication.

Declaration of interests

This research was funded by payments from the National Health and Medical Research Council (Australia) and the Health Research Council (New Zealand) to The George Institute and University of Otago. In addition, CRJ declares consulting fees and honoraria from GSK, AstraZeneca, Chiesi, Sanofi, and Menarini; payment for expert testimony from GSK and Chiesi; meeting or travel support from GSK, AstraZeneca, Chiesi, and Sanofi; participation on data safety monitoring boards or steering committees for LAMA by Night, CICERO–Catalina study Steering committee, COPENICUS trial, VCAPS4, GSK, AstraZeneca, Chiesi, and Sanofi. CRJ is a Director of Lung Foundation Australia, and Director of the Asbestos and Dust Diseases Research Institute Board. RB is a former board member of the Global Initiative for Chronic Obstructive Lung Disease. VMM declares honoraria and meeting or travel assistance from GSK, Boehringer Ingelheim, and Menarini Foundation; and previous grants to her institution from the National Health and Medical Research Council (Australia), Medical Research Futures Fund, and GSK. VMM was a board member of the Thoracic Society of Australia and New Zealand and the COPDX guideline committee. CSF declares consulting fees, honoraria, and meeting or travel support from AstraZeneca, Chiesi, and Sanofi. BC declares research support from GSK; honoraria from Chiesi, AstraZeneca, Inside Practice, Moderna, RX Global, and Lung Foundation Australia; and meeting or travel support from Sanofi and Observational and Pragmatic Research Institute. BC is on the Lung Foundation Australia COPD coordinating committee and the Campbelltown Hospital Drug and Therapeutics committee. HAD declares honoraria from AstraZeneca. RJH declares payments, consulting fees, and honoraria from GSK and AstraZeneca; honoraria from Pharmac (New Zealand); and meeting or travel support from AstraZeneca. RJH is Medical Director of the Asthma Foundation (New Zealand). AM, CLC, IAY, GSH, CPS, CDR, ANA, LB, AD, CG, JPW, CCD, and SG declare no competing interests.

Data sharing

Individual, de-identified participant data will be made available, on request to the corresponding author, to qualified investigators following approval of a protocol that is consistent with the ethics approvals and informed consent obtained in each country and a signed data access agreement with both the trial steering committee and the Research Office of The George Institute for Global Health, Australia.

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